

The Gadfly Project — Custom Web Application Grant 2026

# FCHIP Pilot Build — Community Health Data Platform

*In-kind engineering scope for CHW capture, sync, and dashboards*  
*Your health, our mission.*



FairBanks Community Reach — field operations that need a digital backbone

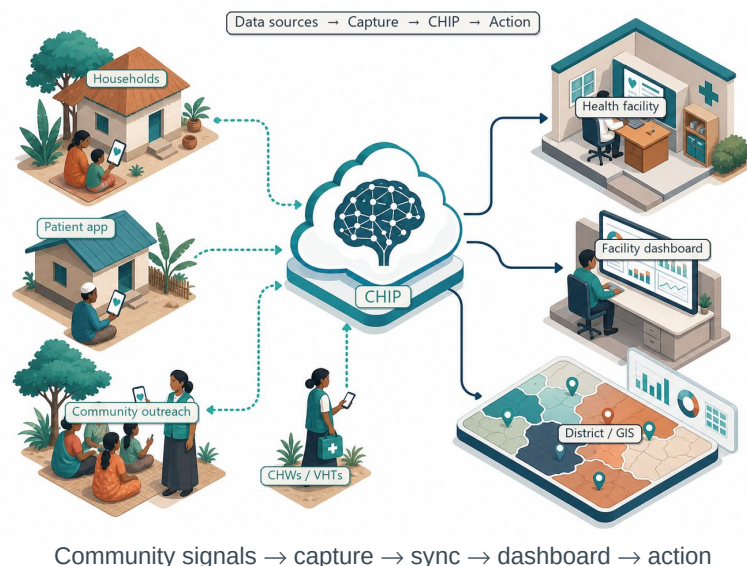
## 1. Engineering overview

FairBanks runs a medical centre and active community outreach across Kampala-area communities. Community Health Workers and Village Health Teams collect vital signs, screening results, and referral notes every week — mostly on paper. That data rarely reaches a shared system in time to prevent outbreaks, flag high-risk pregnancies, or plan medicine stock.

FairBanks Community Health Intelligence Platform (FCHIP) is the digital layer on our community health cascade: offline mobile capture in the field, secure sync to the cloud, and dashboards that turn community signals into early warnings. We are not asking Gadfly for cash. We are asking for professional in-kind development to build the pilot product FairBanks can run, own, and hand to partners.

| Party         | Value                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| For FairBanks | A production-ready pilot: CHW capture app, sync pipeline, and operations dashboard — validated in live outreach, not a slide deck.                           |
| For Gadfly    | A mission-aligned build with real users (CHWs, clinic staff, programme managers), measurable community impact, and a portfolio case in African primary care. |

## 2. Product scope — three modules



### CHW Field Capture

Offline-first mobile web forms for household visits: vitals, maternal flags, immunisation, NCD screening, referrals. Works on low-end Android phones; queues submissions when connectivity drops.

### Sync & Data Layer

Secure API, role-based access, conflict handling, and audit trail from device to cloud. Structured records ready for analytics and export.

### Operations Dashboard

Live view of submissions by area and programme; alert queue for high-risk cases; basic GIS map of activity hotspots; export for reporting.

## 3. Users served in the pilot



CHWs collect data in the field today

- 40+ Community Health Workers and VHTs in Bukoto, Kyebando, Kisaasi, Kamwokya, Kikaaya
- FairBanks clinic nurses and outreach coordinators who triage referrals
- Programme managers tracking maternal, child, geriatric (Gericare), and NCD outreach
- Future district and NGO partners who need interoperable community data feeds

## 4. Technical architecture

| Layer        | Approach                                            |
|--------------|-----------------------------------------------------|
| Presentation | Responsive mobile web (PWA-style) for CHW devices   |
| Offline      | Local queue + retry sync when network returns       |
| Backend      | REST API, PostgreSQL, role-based auth               |
| Dashboard    | Web admin for alerts, maps, and CSV export          |
| Hosting      | Documented deploy; FairBanks-operated cloud account |



Operations dashboard

## 5. Delivery phases

| Phase           | Timing      | Deliverables                                                |
|-----------------|-------------|-------------------------------------------------------------|
| Discovery & UX  | Weeks 1–3   | Field shadowing with CHWs; form design; acceptance criteria |
| Core build      | Weeks 4–10  | Capture app, API, sync, dashboard MVP                       |
| Pilot hardening | Weeks 11–14 | Bug fixes, performance, training materials                  |
| Handover        | Week 15+    | Source repo, docs, admin guide, knowledge transfer sessions |

## 6. Handover & ownership

FairBanks must own and operate this system after Gadfly's build.

- Full source code in FairBanks-owned repository with clear README and environment setup
- Architecture diagram and API documentation for future developers
- Admin and CHW user guides (simple language, screenshots)
- Two live knowledge-transfer sessions with FairBanks technical lead
- 30-day post-handover support window for critical defects

## 7. Impact metrics

- Active CHW users submitting at least weekly during pilot
- Median sync time from field submission to dashboard visibility
- Referrals triggered from flagged high-risk captures
- Community members reached through structured digital records (not paper-only)

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